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Reasoning-agent-driven process simulation, optimization, carbon accounting and decarbonization of distillation



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Distillation is the most energy-consuming unit operation of the chemical industry, however, its decarbonization strategy necessitates laborious manual process simulation, optimization and carbon emission accounting. Here we established a reasoning agent consisting of a large language model (LLM) and an extensive tool set to automate learning material collection, process simulation, optimization and carbon emission accounting of a representative methanol and ethanol distillation case study. Then the agent automatically constructed a heat pump-assisted distillation process to save energy. The impact of three energy supply scenarios on the carbon emissions of distillation, namely, coal, natural gas and renewables, was evaluated. Combining the heat pump-assisted process and renewables could substantially reduce the carbon emission by 98% compared with the coal-based traditional distillation process. This study explored using reasoning agents to automate carbon emission and decarbonization intervention quantification, and facilitated high-resolution carbon emission models of the industry.

The chemical sector accounts for approximately 4% of the global greenhouse gas (GHG) emissions^{1,2}, and its decarbonization requires a high-resolution model to reveal the current emission status across the supply chain³. Levi and Cullen employed material flow analysis to display the global supply chain of chemicals from raw materials to final chemical products⁴. Similarly, Gao and Serrenho visualized the global supply chain of synthetic nitrogen fertilizers, and estimated the GHG emissions from each life cycle stage⁵, which enabled the quantification of mitigation interventions. However, modeling the supply chain of numerous chemicals, relevant production technologies, and GHG emissions faces several challenges, such as the overwhelming workload and the professional knowledge barrier for each process. Furthermore, it is an interdisciplinary effort to look for cost-efficient and mature decarbonization technologies. Taking distillation – a widely used separation technology and energy-intensive unit operation⁶ – as an example, a heat pump is proposed as an energy-saving technology to make full use of the latent heat of the vapor by transferring it to the reboiler,

using a compressor to increase the vapor pressure and enable a condensation temperature higher than the reboiler temperature⁷. Yet, this tedious process of simulation and optimization is currently conducted by practitioners with expertise in both chemical engineering⁸ and carbon emission accounting⁹. Thus, it is vitally important to develop new technologies to tackle the interdisciplinary task and accelerate GHG accounting and decarbonization planning.

Fortunately, the development of artificial intelligence (AI) brings new opportunities to address this issue¹⁰. Unlike task-specific machine learning (ML) models such as AlphaGo and Alphafold^{11,12}, LLMs based on Transformer can process natural language and have rich knowledge of science and engineering¹³. On top of that, chain-of-thought technology enables high-quality reasoning and programming¹⁴. External tools such as memory, search engines, and professional software can be connected to LLMs to establish agents that perform functions based on autonomous reasoning. Parallel computing can accelerate the modeling process, and human experts

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can focus on goals, workflow, standards, result verification, and suggestions to ensure model quality, thanks to the interpretability of LLMs' outputs. Thus, it is promising that agents based on LLMs can be tailored for process design and GHG accounting, which is the focus of the study.

To quantify the GHG emissions of chemical and energy process systems, life cycle inventory can be sourced from databases (e.g., Ecoinvent), published literature, industrial datasets, and accurate process simulation¹⁵. Process simulations are typically carried out using commercial tools such as Aspen Plus, HYSYS, and gPROMS. For instance, Hao et al. developed a hybrid modeling framework combining first-principles simulation with ML to assess an integrated power plant with carbon capture and utilization technologies¹⁶. In the work of Hao et al., commercial simulators were used to generate detailed input-output data for sub-systems, which were then approximated using ML-based surrogate models. These surrogates were subsequently assembled to perform techno-economic analysis and life cycle assessment at the system level. Other researchers have similarly leveraged ML in conjunction with process simulators to accelerate system-level assessment. For example, Shi et al. used Aspen Plus to model biomass gasification in a bubbling fluidized bed, and adopted ML to predict fast pyrolysis products¹⁷. Timilsina et al. combined ML with Aspen Plus to simulate methanol synthesis, enabling process optimization and economic viability assessment for waste-to-methanol production¹⁸. These approaches reduce computational demands while enabling broader design space exploration, but ML algorithms suffer from low interpretability and reliance on relevant time- and resource-intensive manual training datasets. This presents a scalability challenge, especially when attempting to model supply chains or regional-scale energy systems, where the number of interconnected units and decision variables increases dramatically. Unlike conventional ML models, LLMs could explicitly mimic experts' reasoning process with few-shot prompting and have been gradually adopted since their release. Pan et al. adopted LLMs to extract information from the literature¹⁹, and Ruan et al. used LLM-driven agents to design experiments and analyze results²⁰. Although LLMs have been applied to generate process flowsheets²¹ and conduct process optimization²², few studies have explored their application in integrating process simulations with GHG emission calculation. This gap presents a compelling research opportunity to combine recent advances in generative AI with traditional chemical engineering workflows, potentially enabling more scalable and accessible sustainability assessments.

To fill this gap, we have developed an agent capable of chemical process simulation, optimization, and GHG accounting. A distillation column to separate methanol and ethanol was used as a proof-of-concept case study for solvent recovery²³, and a heat pump-assisted distillation process was simulated and optimized to save energy. Then, the carbon emissions using different fuels were evaluated to inform the decarbonization strategy.

Results

We have developed an agent capable of reasoning and utilizing an extensive suite of tools. With the prompts of natural language commands, the agent can autonomously execute process simulation, optimization, carbon accounting, and result visualization, with all results rigorously validated by the authors to ensure reliability.

Simulation and optimization of the traditional distillation process

A reasoning agent was first established by combining an LLM, Grok 3, and several tools, including Aspen Plus, augmented-retrieval generation (RAG), and so on, as shown in Fig. 1a.

The agent followed the workflow defined in Fig. 1b to automatically design a distillation column to separate a 100 kg h⁻¹ feed mixture containing 40 wt% (weight percent) methanol and 60 wt% ethanol into a top stream with 98 wt% methanol and a bottom stream with 98 wt% ethanol.

The agent firstly searched for example Aspen Plus backup (.bkp) files on the Internet and downloaded 131 .bkp files, among which 125 files could be executed by local Aspen Plus. The Aspen files were used, manually verified to ensure broad coverage across process blocks, component types,

and physical property methods, providing sufficient diversity for the agent to generalize and generate valid input files within the current project scope. To provide parameters for the rigorous distillation model (RadFrac), the agent generated an input (.inp) file for a shortcut distillation column (DSTWU) model by combining the official input language guide²⁴ and 20 examples relevant to DSTWU and components. The agent selected the non-random two-liquid (NRTL) method to calculate the activity coefficient for the distillation system. The DSTWU model estimated that the minimum reflux ratio, R_{min} , and total number of theoretical plates to realize the separation task were 3.08 and 16 (Table S1 in the Supplementary Information), respectively, and the top and bottom flows were calculated as 39.6 kg h⁻¹ and 60.4 kg h⁻¹, respectively.

These results were fed to the agent along with the official guide²⁴ and 13 relevant examples to generate the .inp file for the rigorous RadFrac model (Fig. 2a). Given that the top and bottom streams, D and W , had been calculated by the DSTWU model, the number of theoretical stages, N_t , feed stream location, F_s , and reflux ratio needed to be optimized to minimize the heat duty, consequent operation energy cost and GHG emissions from the distillation system.

A sensitivity analysis and a design specification were designed by the agent to execute the process optimization. The agent set various combinations of N_t and F_s in a sensitivity analysis. Since the reflux ratio was the only design degree of freedom²⁵ in the steady state distillation column design when other parameters, such as pressure, F , and D , were all set, the design specification adjusted the continuous reflux ratio to reach the target top concentration for each combination of N_t and F_s .

Figure 2b shows a common trend that as the number of theoretical stages increased from 21 to 40, the reflux ratio dropped. Consequently, the reboiler heat duty decreased as shown in Fig. 2c. For each N_t , a best feed stream location, F_s , could be found to minimize the reflux ratio and heat duty to accomplish the separation task. To be specific, as N_t increased from 21 to 40, the best F_s shifted from 12 to 23, and the reflux ratio decreased from 6.77 to 2.76 accordingly. Thus, the heat duty was reduced from 98.6 kW to 50.3 kW.

The reflux ratio would approach the minimum reflux ratio when N_t is further increased. Considering that the minimum reflux ratio was estimated by the DSTWU model with several assumptions, a sensitivity analysis was set by the agent to identify the actual minimum reflux ratio using the rigorous RadFrac model, varying N_t from 90 to 320. As shown in Fig. S1, the lowest reflux ratio for each N_t decreased and then levelled off as N_t increased, revealing that the actual minimum reflux ratio of this case was 2.52. However, Luyben proposed that the actual reflux ratio should sustain a minimum threshold of 1.2 times the theoretically established minimum reflux ratio in principle²⁵, which was 3.02 in this case study. Considering the criterion is intended as a rule of thumb rather than a rigid specification, the optimal design that had 34 theoretical stages with the feed stream located above stage 20 was selected because the reflux ratio was the closest to the recommended threshold. As shown in Table S1, the design resulted in a minimized reboiler duty of 53.7 kW, and a reflux ratio of 3.04.

GHG emission calculation and visualization

The optimal operating condition had minimized the energy demand for the distillation system. However, distillation is known as the most energy-intensive unit operations⁶, implying considerable GHG emissions from steam and electricity generation. It is important to quantify the relevant GHG emissions and evaluate the decarbonization strategy.

The agent was required to establish three energy supply scenarios to evaluate the impact of different fuels on the GHG emissions. The coal scenario used coal-based steam and electricity to provide energy to the system, and the natural gas scenario assumed natural gas-based steam and electricity. The renewables scenario adopted biomass for steam generation and wind power.

The agent extracted the steam demand from the simulation results and calculated the electricity demand to circulate the cooling water and drive the fan of the cooling tower, as detailed in the "Methods" section. Next, the agent

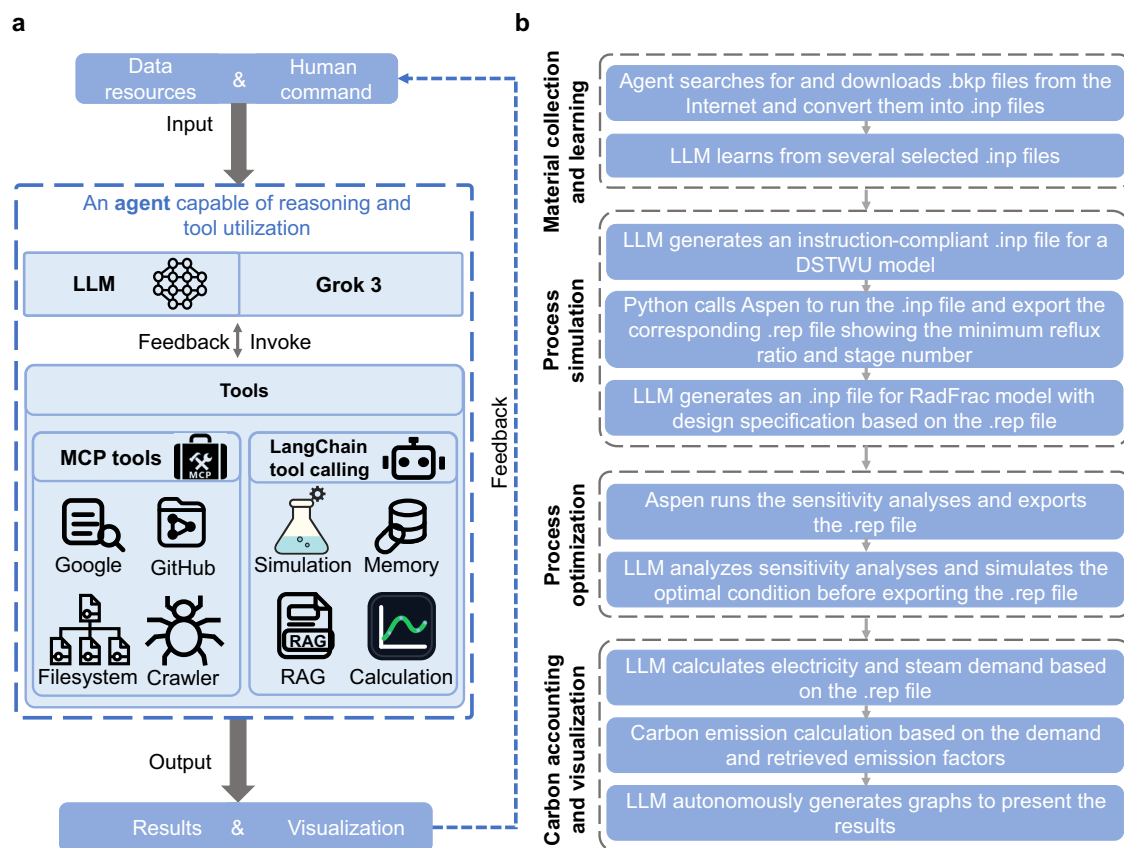


Fig. 1 | The agent autonomously simulates a distillation process and calculates the carbon emissions. **a** Framework of the agent consisting of a reasoning LLM and tools. **b** The AI-driven workflow autonomously executes process simulation, optimization, and carbon emission accounting. LLM stands for a large language model.

MCP represents the model context protocol. RAG is an abbreviation for retrieval-augmented generation. .bcp, .inp, and .rep are backup, input, and report files for Aspen Plus, respectively; DSTWU and RadFrac stand for shortcut distillation design and rigorous distillation, respectively.

automatically searched the RAG database for corresponding emission factors and their ranges^{26–28}, to calculate the GHG emissions from energy consumption of the distillation system. As shown in Fig. 3, the coal-based scenario had the highest GHG emissions of 20.3 kg CO₂e h^{−1}, while the total emission would drop by 44% and 93% in the natural gas and renewable scenarios, respectively. In the first three scenarios, more than 95% of the emissions arose from steam generation, highlighting the necessity to decarbonize steam generation. Since biomass absorbs CO₂ during growth, the CO₂ emissions from biomass combustion were neglected in current GHG emission accounting practices²⁹. The emissions from the renewables scenario were attributed to the harvesting, processing, and transport of biomass²⁶.

Heat pump-assisted distillation process for energy saving and decarbonization

The comparison of the first three scenarios in Fig. 3 assumed constant energy demand by distillation, while energy-saving technologies could be developed to reduce the GHG emissions. One example is heat pump-assisted distillation columns, which pressurize the vapor from the top of the column to reach a high condensation temperature and provide heat to the reboilers using the latent heat^{25,26}. It could avoid using steam and subsequently cooling water operation cost at the cost of compressor work driven by electricity.

To quantify the potential energy-saving and decarbonization effect of a heat pump, the agent was asked to convert the optimal traditional distillation process to a heat pump-assisted process by generating an .inp file for the process displayed in Fig. 4. 10.7 kW of electricity was required to increase the vapour pressure and saved 53.7 kW of steam input to the reboiler. Consequently, the heat pump-assisted process could save 80% energy demand of

the optimized traditional distillation process. The GHG emissions (Fig. 3) were reduced to 8.92, 5.69, and 0.34 kg CO₂e h^{−1} when coal, natural gas, and wind power were used, respectively. When coal was the energy source, the heat pump-assisted process could reduce GHG emissions by 56% compared with the optimal traditional distillation process.

Discussion

The time-consuming quantification of current emission status and decarbonization strategy is a prerequisite for decarbonizing the chemical sector. To overcome the laborious workload, we have designed a reasoning agent that could automate chemical process simulation, optimization, and carbon emission accounting for a representative methanol and ethanol separation case study. The model indicates that the GHG emissions reached the highest value of 20.3 kg CO₂e h^{−1} when coal was used to provide steam and electricity, while combining a heat pump and renewable electricity could reduce the carbon emissions by 98%.

The agent exhibits several advantages for automatic process simulation, optimization, and carbon emission accounting. Firstly, the transparency of the process enables interpretability and verification, which could be widely adopted by practitioners with trust. Secondly, the developed workflow could be transferred to other chemical process designs, such as multi-component separation and even other process industries. Thirdly, the inherently parallel nature of the agent has enabled large-scale deployment, although each run took around 4 h for this case.

The state-of-the-art of carbon emission accounting for the chemical sector combines high-resolution material flow analysis with life cycle assessment^{3,5,30}, however, not all the processes and products can be covered due to the overwhelming workload. The developed agent is expected to contribute to high-resolution models, accelerate the carbon emission

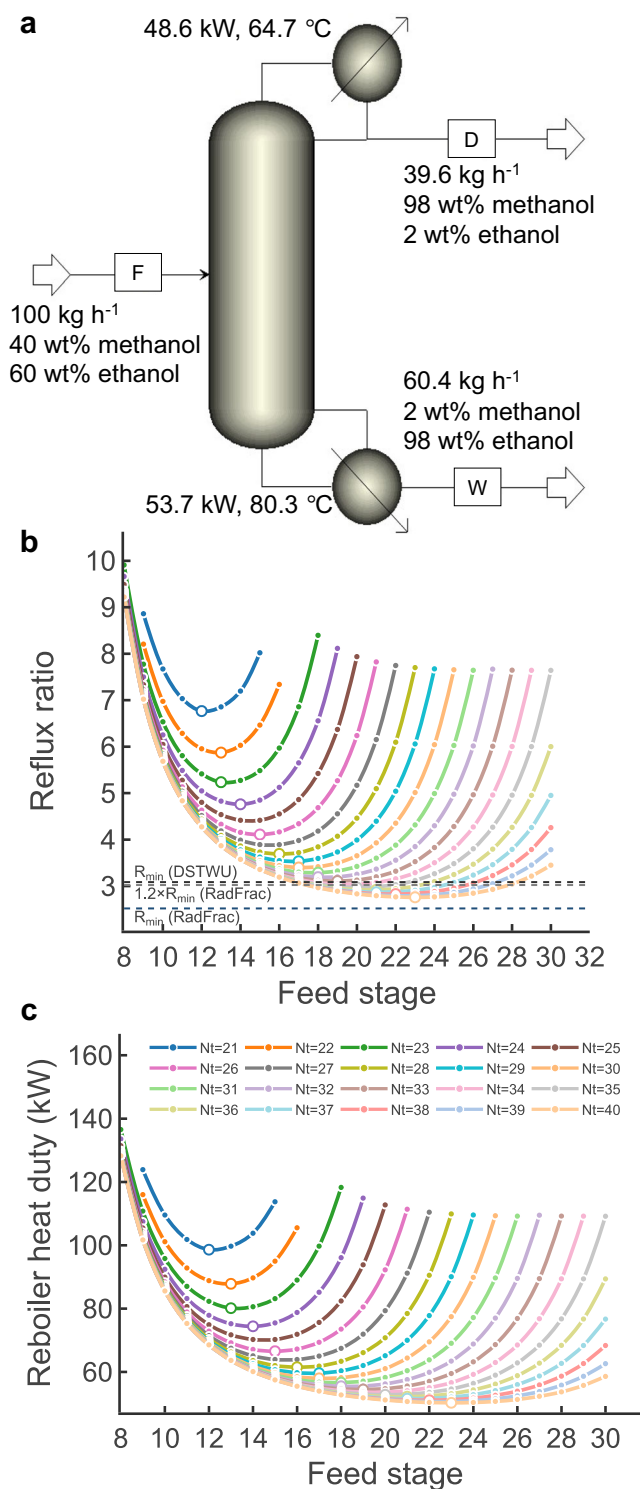


Fig. 2 | Impacts of the theoretical stage number of the rigorous distillation model, N_t , and feed stage on the reflux ratio and reboiler heat duty. a The rigorous distillation model. **b** The effect of N_t and feed stage on the reflux ratio; **c** The effect of N_t and feed stage on the reboiler heat duty. The hollow circles represent the best feed stage for each number of theoretical plates to obtain the lowest reflux ratio and heat

duty. wt% stands for weight percent. Simulation condition: feed steam of 100 kg h⁻¹ 40 wt% methanol and 60 wt% ethanol, top pressure 101.3 kPa, pressure drop 10 kPa, methanol concentration in the top stream 98 wt%, and ethanol concentration in the bottom stream 98 wt%. (a) and (b) were drawn by the code generated by the agent according to the results.

accounting of the chemical industry, and facilitate the systematic quantification of decarbonization strategies.

The adopted LLM, i.e., Grok 3, exhibits a high reasoning capability to successfully generate the .inp files and conduct carbon emission accounting that needs complex multi-stage reasoning. The developed agent served as a prototype that successfully integrated reasoning with the Reasoning-Acting

framework, with its current efficacy being shaped by both prompts and the selection of LLMs. The workflow was pre-defined in the prompts for this case study to demonstrate the capability of automatic process design and carbon accounting, while tackling complex problems exceeding the input and output token limits of LLMs would require autonomous agentic task-decomposition and step-by-step execution³¹. Progressively increasing the

autonomy and generative power of each module represents a key direction for future work.

Given the 20–30 year lifespan of chemical plants, cumulative operational emissions from energy and feedstock use far exceed the one-time embodied emissions of equipment. The embodied emissions associated with equipment manufacture and installation are assumed to be negligible in comparison with operational emissions in the current analysis. This approach is consistent with life cycle assessment practice in the chemical industry, where equipment manufacturing emissions are seldom treated as the primary hotspot^{32,33}.

The heat pump-assisted distillation process could substantially reduce the energy demand by 80%, so the operation cost and total cost are also lower than the traditional distillation process³⁴, despite the capital expenditures on the compressor. This is a good example of simultaneous energy, GHG emission, and cost reduction. Shifting to wind power from coal-based electricity could reduce the emission factor and the total emissions, but ensuring a stable supply of wind power remains a challenge for the current infrastructure, which requires coupling thermo storage³⁵ or complements of fossil-fuel-based electricity. Methanol vapour is superheated after compression in this case study; however, a wet fluid, e.g., other vapor streams, could condensate and damage the compressor without preheating. The temperature-entropy diagram can be used in the design phase to distinguish if a fluid is dry or wet, with the latter one requiring preheating³⁶.

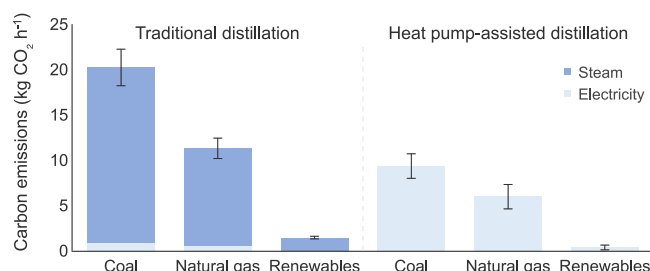
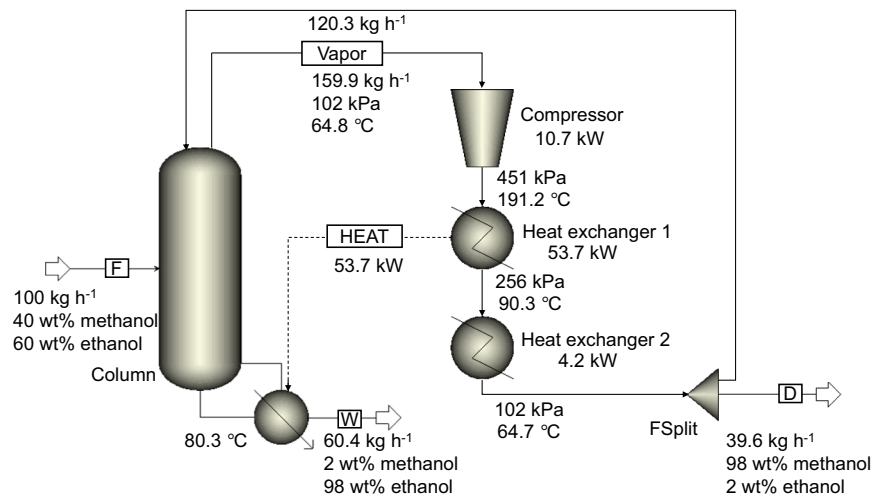


Fig. 3 | The impact of fuel type on the GHG emissions from the optimal distillation column design and heat pump-assisted process. The error bars represent the emissions calculated using the upper and lower bounds of the GHG emission factors in each scenario. Design of the optimal distillation column: feed steam of 100 kg h⁻¹ 40 wt% methanol and 60 wt% ethanol, top pressure 101.3 kPa, pressure drop 10 kPa, methanol concentration in the top stream 98 wt%, and ethanol concentration in the bottom stream 98 wt%, number of theoretical stages 34, feed stream location 20, reflux ratio 3.04. The figure was drawn by the code generated by the agent according to the results.

Fig. 4 | The heat pump-assisted distillation process. Simulation condition: feed steam of 100 kg h⁻¹ 40 wt% methanol and 60 wt% ethanol, column top pressure 102 kPa, pressure drop 9.8 kPa, methanol concentration of 98 wt% in the top stream, and ethanol concentration of 98 wt% in the bottom stream. A design specification was used to adjust the compressor outlet pressure to match the heat duties of the reboiler and Heat exchanger 1. wt% stands for weight percent.



Methods

We developed an agent capable of reasoning and tool utilization, as detailed in the first subsection. The following subsections present a representative workflow in which the agent autonomously acquires the necessary learning materials, executes the process simulation, performs optimization, and finalizes the corresponding carbon emission assessment.

Agent establishment

As shown in Fig. 1a, Grok 3 acted as the central large language model, interpreting natural language prompts, invoking the requisite tools, and iteratively refining its actions on the basis of tool feedback, such as Model Context Protocol (MCP), for operations like web searching, repository inspection, and file transfer, until the desired output was generated and visualized.

The agent was implemented with LangChain, which mediates all interactions between Grok 3 and external software modules. Two classes of auxiliary tools were exposed to the model: (i) LangChain-based tool calling and (ii) MCP tools. LangChain also maintained a short-term conversation memory that stored prior prompts and tool responses, enabling Grok 3 to condition each new action on the cumulative dialog state.

LLM-driven learning material collection via MCP

We set up an MCP-based workflow that links the model to external utilities for web searching, repository inspection, and file transfer. Retrieved exemplary Aspen Plus files would be provided to the agent. Because auto-saved backup (.bkp) files are more widely available online than .inp files, the agent located and downloaded 131.bkp files and then converted 125 valid cases of them to human-readable.inp format for subsequent simulations. MCP, an open standard connecting LLMs to third-party tools and resources, enabled seamless task execution through a single, audible interface, extending the agent's capabilities beyond text generation while maintaining oversight and security. To facilitate this process, we configured MCP with several key tools that the LLM could invoke as needed. The key tools included GitHub access, file system access, a web search engine, and web crawler utilities. Collectively, this constitutes a Reasoning-Acting framework³⁷, where the model leveraged a chain-of-thought to break down complex tasks. It then iterated through steps of reasoning and action, pausing after each act to observe the new state before the next reason, thereby enhancing its problem-solving capabilities.

With these utilities, we developed targeted prompts that guided the agent to .bkp files. The prompts provided domain context, including expected file extensions, typical naming conventions, and common hosting sites, and it explicitly instructed the agent to ignore irrelevant hits such as forum requests or invalid links. This prompt engineering supplied clear domain-specific guidance and substantially reduced false positives during

retrieval. Note that all prompts and responses of the agent are shown in the code repository and Supplementary Information.

The overall retrieval task was decomposed into a sequence of subtasks that the agent executed stepwise, aligning with established strategies for complex task handling by AI agents. The process comprised four main steps:

- 1) Locating potential resources: The agent identified candidate sources of .bkp files via GitHub and web search. The output of this step was a list of potential URLs or repository links, each with a brief description.
- 2) Accessing resource content: Each resource was then accessed using the appropriate tool—either the crawler for webpages or the GitHub tool for repositories. The agent gathered textual or directory content for further analysis.
- 3) Evaluating content for download links: The LLM analyzed the retrieved content to identify valid and accessible .bkp file links, filtering out noise and broken resources.
- 4) Downloading selected files: Using the compiled list of verified links, the agent invoked file-handling operations to batch-download the files to the local file system.

To accommodate the LLM's context length limitations, the workflow was designed to externalize intermediate results and execute iteratively:

- (1) Each located resource was recorded in an individual markdown file containing a link and a short description.
- (2) For each resource, content retrieval was executed separately and saved to its own markdown file.
- (3) Each markdown file was then independently evaluated to extract downloadable .bkp links.
- (4) A batch-download operation was performed using the aggregated list of verified links.

Process simulation

The agent firstly adopted the shortcut distillation design model using the Winn-Underwood-Gilliland method (DSTWU) to estimate the minimum reflux ratio²⁵, number of theoretical stages, and feed location, then used the rigorous distillation model (RadFrac) to conduct the accurate simulation.

We adopted two methods to control Aspen Plus: the .inp file and the application programming interface (API) command. An .inp file defines all input configurations for the simulation, such as components, physical property method, flowsheet, block, and steam parameters, serving as the command script for Aspen Plus execution. It contains a compact and human-readable text-based summary of the specifications that the agent could automatically generate .inp files based on prompts, selected simulation examples, and the official input language guide²⁴. The agent could also control Aspen Plus to import and export .inp, .bkp, and .rep files through API commands. Note that .bkp files contain complete simulation information in non-human-readable binary format, and .rep files provide a comprehensive summary of the key results and data generated from the simulation, facilitating output analysis.

We wrote a natural language prompt shown in the SI and fed 20 .inp files relevant to DSTWU and components, along with the official input language guide²⁴, to the LLM to set up the separation task. Thus, an .inp file for the DSTWU model could be generated. A 100 kg h⁻¹ flow of methanol (40 wt%) and ethanol (60 wt%) needed to be separated by atmospheric distillation. The concentration at both the top and bottom should be 98 wt% for methanol and ethanol, respectively. Since the recovery rates of both the light and heavy components, RR_L and RR_H , in Eqs. (1) and (2), were the required input values for the DSTWU model, we asked the LLM to calculate the mass balance based on Eqs. (3) and (4) and subsequently RR_L and RR_H . The agent was also required to select the proper physical property method for the system. After the agent saved the generated .inp file, the agent called Aspen Plus to import the .inp file, run the simulation, and export the results

through a .rep file.

$$RR_L = \frac{D \times x_D}{F \times x_F} \quad (1)$$

$$RR_H = \frac{D \times (1 - x_D)}{F \times (1 - x_F)} \quad (2)$$

$$F = D + W \quad (3)$$

$$F \times x_F = D \times x_D + W \times x_W \quad (4)$$

where, F , D , and W are the mass flows of the feed, top, and bottom streams, in kg h⁻¹, respectively; x_F , x_D , and x_W represent the mass concentration of the light component, methanol in this case, in the feed, top, and bottom streams, respectively.

The agent read the .rep file and obtained the minimum reflux ratio, feed location, and number of theoretical stages, N_t . Then it used the estimated reflux ratio, number of theoretical stages, and feed stage location, F_s , and generated an .inp file to conduct the rigorous distillation simulation using RadFrac. Since RadFrac is a typical operating calculation where equipment parameters and operating conditions are input variables while the separation effect is to be calculated, the agent adopted a design specification to reach the target product concentrations. As shown in the .inp file shown in the SI, the reflux ratio was adjusted to reach 98 wt% methanol in the top stream for a given combination of number of theoretical stages and feed location.

Process optimization

We further asked the agent by prompts to optimize the number of theoretical stages and feed stage location by a sensitivity analysis suitable for discrete variables and a design specification for the continuous variable to reach the lowest reboiler heat duty and GHG emissions. Given that the two variables are discrete, the generated .inp file contains a group of sensitivity analyses. The number of theoretical stages and the feed stage location were varied over a wide range at the discretion of the LLM. For each combination of the number of theoretical stages and the feed stage location, the design specification was conducted to obtain the optimal reflux ratio, a continuous variable, and heat duty to complete the separation task. The overall process thus constitutes an optimization involving both discrete and continuous variables.

Heat pump-assisted distillation process

In the traditional distillation process, heat is provided by steam to the reboiler, and cooling water removes the condensation heat of the vapour in the condenser. To utilize the condensation heat, a heat pump-assisted distillation process was designed by the agent.

The LLM was fed with several elements, including prompts describing the process, the official input language guide, the .rep file showing the results of the optimal RadFrac model, and selected example .inp files with relevant blocks. Then, the .inp file for the heat pump-assisted process would be generated by the LLM.

The prompts (displayed in the Supplementary Information) indicated that the parameters of the columns were the same as the RadFrac model, except for the absence of a condenser. The vapor leaving the column was compressed with a compression ratio of 5, before it condensed in Heat exchanger 1 and released the latent heat through a heat stream to the distillation column reboiler. The expert-provided isentropic efficiency of 0.72³⁸ and mechanical efficiency of 0.85^{26,39} were adopted by the agent. To fully use the latent heat, a vapor fraction of 1 and an outlet temperature 10 °C higher³⁵ than the reboiler temperature of 80.3 °C were set for Heat exchanger 1. Then the pressurized saturated liquid was further cooled down to fit the pressure and temperature at the column top using cooling water in Heat exchanger 2. A stream splitter was used to divide the condensate to reflux

and top product, *D*. A design specification was designed by the LLM, which adjusted the compressor outlet pressure to match the heat duty of Heat exchanger 1 and the reboiler (Fig. 4), to reduce the compression work. Then the agent called Aspen Plus to run the .inp file and export the simulation results for further analysis.

Carbon emission accounting and results visualization

The agent adopted the simulation results of the traditional distillation process and heat pump-assisted process to calculate the steam and electricity demand for the quantification of GHG emissions.

Two sources of energy demand related to GHG emissions: steam generation and electricity for driving the pump of cooling water, the fan of the cooling tower, and the compressor. As shown in Eq. (5), the total GHG emission, GHG_t , in $\text{kg CO}_2\text{e h}^{-1}$, was calculated by steam and electricity's emission factors and their consumptions.

$$GHG_t = \left(\text{Consump}_{\text{steam}} \times EF_{\text{steam}} + \text{Consump}_{\text{electricity}} \times EF_{\text{electricity}} \right) \times 3600 \quad (5)$$

where $\text{Consump}_{\text{steam}}$ and $\text{Consump}_{\text{electricity}}$ are the consumptions of steam and electricity, in kJ s^{-1} , respectively, and EF_{steam} and $EF_{\text{electricity}}$ are the emission factors of steam and electricity, in $\text{kg CO}_2\text{e kJ}^{-1}$, respectively.

The required heat duty of the reboiler provided by the steam was calculated by Aspen Plus and used directly by the LLM.

The electricity demand to drive the cooling water for the condenser was calculated by the agent as the following procedure. Firstly, the heat duties of the condenser and Heat exchanger 2 were used to calculate the flow rates of cooling water, in kg h^{-1} , assuming a 5°C temperature rise, TR , as Eq. (6). Then the agent estimated the pipe diameter using an average velocity of 2 m s^{-1} and then rounded to a commonly used pipe diameter before it calculated the final average velocity. To estimate the energy input to the cooling water, W_e , the parameters in Eq. (7) were estimated by the agent as shown in the Supplementary Information. Then the electricity demand of the pump was calculated using Eq. (8).

$$\text{Flowrate} = \text{HeatDuty}_{\text{cooling}} / \text{HeatCap} / TR \quad (6)$$

$$\frac{W_e}{g} + Z_1 + \frac{u_1^2}{2g} + \frac{P_1}{g} = Z_2 + \frac{u_2^2}{2g} + \frac{P_2}{g} + \frac{\sum h_f}{g} \quad (7)$$

$$\text{Consump}_{\text{Pumpelectricity}} = \frac{W_e \times \text{Flowrate}}{\eta} \quad (8)$$

where $\text{HeatDuty}_{\text{cooling}}$ is the cooling duty of the condenser or Heat exchanger 2; HeatCap is the heat capacity of cooling water; W_e stands for the energy input into the cooling water, in J kg^{-1} ; Z_1 and Z_2 are the heights of the pump inlet and the condenser, and the latter one was estimated by the LLM according to the number of theoretical stages; u_1 and u_2 are the average velocity at the pump inlet and the condenser; P_1 and P_2 are the pressure at the pump inlet and the condenser; $\sum h_f$ stands for the total flow resistance as detailed in the SI; g is the gravitational constant; η represents the typical efficiency of the cooling water pump of 0.6⁴⁰.

To obtain the electricity demand by the fan of the cooling tower, a product manual was given to the agent to select the proper cooling tower aligned with the required heat duty and flow rate, as shown in the Supplementary Information. As to Heat exchanger 2 in the heat pump-assisted distillation process, the cooling water flow rate was too small for a cooling tower, so we used prompts to ask the LLM to calculate the fraction of cooling water flow rate in the total flow rate and estimate the corresponding electricity consumption.

Regarding the emission factors, we asked the agent to evaluate three scenarios: steam and electricity generation by three kinds of fuels, namely, coal, natural gas, and renewables, to reveal the impact of fuel type on the

process GHG emissions. To retrieve the emission factors and their ranges for each scenario, we built an RAG system. The emission factors used in subsequent calculations were verified and provided in the Supplementary Information.

Carbon emission factors are often buried within a vast and rapidly growing body of scientific literature, government reports, and technical documents. The manual process of searching, downloading, reading, and extracting specific carbon emission factors is time-consuming. By combining the automated literature retrieval technique with RAG, we have developed a system that is capable of searching for relevant online sources (academic databases, institutional websites) for literature potentially containing desired emission factors, downloading accessible documents, and extracting specific carbon emission factors using a RAG system that leverages the power of LLMs informed by the retrieved document content. In this way, RAG enables the LLM to provide specific, contextually relevant responses based on the provided literature, thereby improving accuracy and reducing hallucinations⁴¹.

RAG operated in the following manner: The initial step was to index relevant documents (100+ downloaded literature), which included parsing their content into a machine-readable Markdown format; chunking these documents into smaller, manageable text segments; converting each chunk into a numerical vector representation using the bge-base-en-v1.5 embedding model; and storing these embeddings in a specialized vector database by Qdrant. Retrieval was the second step. When a user asked a question (e.g., What is the exact value of default, lower, and upper CO_2 emission factors for natural gas combustion?), the question was embedded, and the vector database was searched for text chunks whose embeddings were most similar to the question's embedding. The final stage was generation. The original question and the content of the retrieved text chunks were passed together as context to the LLM, i.e., Grok 3. Official data sources such as the IPCC Guidelines for National Greenhouse Gas Inventories²⁹ were prioritized over other data sources. The LLM then generated a response based specifically on the provided context from the documents, rather than relying solely on its general knowledge.

Sensitivity analyses were conducted to reveal the impact of energy emission factors. The upper and lower bounds of the steam generation emission factors were used to calculate the range of GHG emissions of each scenario of traditional distillation, considering the dominating contribution of steam. The upper and lower bounds of electricity emission factors were used to calculate the emissions for the heat pump-assisted distillation process. After the carbon emissions in all the scenarios were evaluated, the results and prompts were passed to the agent to generate figures in.html format.

Data availability

The data and prompts used for the study were all disclosed in the paper and the Supplementary data.

Code availability

The code used to generate the results was available from the following link upon request: <https://github.com/TSH-AI/Aspen-AI>

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Author contributions

S.T. wrote the code and prompts. X.Z. was responsible for the MCP tools. H.Z. built the RAG. Y.X. drew the figures. Z.H., L.C., G.S., and Q.S. contributed to agent establishment. Y.G. wrote the manuscript, acquired funding, supervised, and conceptualized the study. W.W. supervised the project and acquired funding. All the authors revised the manuscript.

Competing interests

The authors declare the following competing interests: X.Z. is hired by PowerChina Huadong Engineering Corporation Co., Ltd., and L.C. is hired by Sinopec Engineering Incorporation.

Additional information

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